Project 1: Data Analysis Project

Objectives. In your data analysis project, you should show that you can

1. download online data using an API;
2. combine and restructure data from two tables;
3. present results in figures and interpret them in economic terms;
4. simulate a population and use it for a counterfactual;
5. organize and document a programming project.

Scope. Complete Parts 1 and 2. The starter notebook supplies a download function and the first query. Everything else uses tools from the lectures and problem sets.

Structure. Present your analysis in one notebook (.ipynb). Write the simulation as a documented function in a Python module (.py) and import it into the notebook. Keep both files together; restart the kernel and run all cells in order before submitting.

Hand-in. First, click here to register your group in the Google Sheet. When your project is ready, click here to upload your group ZIP file to Dropbox. Submit one ZIP file per group. Name it **GroupName_P1_v1.zip**, using the group name from the sheet. You may work individually or in groups of up to four.

Deadline. Sunday, 29 November 2026 at 23:59, for both group registration and Project 1. See the course plan: Projects and Exam.

Feedback. Your TA will provide feedback on your project.

Exam. Your data analysis project forms part of your exam portfolio. You can incorporate the feedback before handing in the final version.

1 Measuring Wealth Inequality in Denmark

How wide would the national wealth gap be if every adult had the typical wealth of their age group? And what happens to Danish wealth if the population gets older? Start by measuring the differences in the data. Net wealth equals assets (such as housing, deposits and pensions) minus liabilities. With assets of DKK 200,000 and debts of DKK 250,000, a person has negative net wealth of $-50{,}000$ DKK.

1.1 Download and Read the Tables

Use Statistics Denmark's FORMUE16 and FORMUE15. In FORMUE16, one row gives a *percentile threshold*: P10 is the wealth level below which roughly 10% of the group falls, and P50 is the median. In FORMUE15, one row gives the *number of people* in a wealth bracket.

1. Run the supplied FORMUE16 query. Adapt it to download all wealth brackets from FORMUE15 for 2024, all age groups, nominal prices and the 2020 wealth definition.
2. Inspect the first rows of each result. Which age code denotes all adults? Set the all-adult rows aside for Part 2 and keep the eight age groups for the age profiles.

Hint. Each keyword in `download(...)` is one variable code, and `'*'` selects all values. This is the same information as the `variables` list in the Lecture 10 API notebook. Open the FORMUE15 table information in your browser to find the variable codes and their values.

1.2 Plot the Age Profiles

1. Draw one figure with age groups along the horizontal axis and three lines for P10, P50 and P90. Label the vertical axis in DKK or million DKK.
2. Within each age group calculate the upper gap (P90 − P50) and the lower gap (P50 − P10). Which age group has the widest upper gap? Give the amount.
3. In which age groups is P10 negative? Give one economic reason why young adults are more likely than older adults to have more debt than assets.

Hint. Turn the percentile codes into numbers (`'P10'` becomes `10`) with `.str.replace` and `.astype(int)`, as in Lecture 10. Then use `pivot_table` (Lecture 9) to get one row per age group and one column per percentile. The data are one cross-section: you compare different people of different ages in 2024, not the same people as they age.

1.3 Who Has Negative Net Wealth?

Within each age group, divide the number of people in the brackets below zero by the total across all brackets, and express the result as a percentage. Merge the counts and percentages with your percentile table by age. Plot the percentage by age and describe one contrast with the median profile. Check that the eight age counts sum to the published all-adult count.

Hint. Three brackets lie below zero; identify them from the bracket labels in the table information. `groupby` sums counts by age, and `pd.merge(..., validate='1:1')` checks that each age group appears once (Lecture 10).

2 Simulating the Wealth Distribution

You now build artificial economies of 10,000 adults. In Sections 2.1 and 2.2, keep each simulated adult's age group fixed and change how you assign wealth. Put the simulation in a documented function in a separate module and import it into your notebook. Let the function take the age shares as an argument: you will change them in Section 2.4.

2.1 Give Everybody Their Age-Group Median

1. Divide each age count by the total across the eight age groups. These shares are the probabilities of belonging to each group. Check that they sum to one.
2. Draw an age group for each adult. Draw a number u from the uniform distribution on $[0, 1)$. Split $[0, 1)$ into eight intervals whose lengths equal the age shares: the first group covers $[0, s_1)$, the second $[s_1, s_1 + s_2)$, and so on. The adult belongs to the group whose interval contains u.
3. Give each simulated adult the P50 wealth of their group. Compare the simulated age shares with the published shares. Why do they differ slightly?

Hint. Use `rng = np.random.default_rng(42)` and `rng.uniform(0, 1, size=10000)` as in Lecture 8. Loop over the eight groups, keep a running sum of the shares for the interval bounds, and select adults with a Boolean mask such as `(u >= lower) & (u < upper)`. `np.mean(age == j)` is the simulated share of group `j`. Keep the shares and the wealth data in the same age order.

2.2 Allow Wealth to Vary Within Each Age Group

Keep each person's age group. Draw a rank from 1 to 99 for each adult, with each rank equally likely. Assign the published wealth threshold for that person's age group and rank. For example, a person aged 40–49 who draws rank 25 receives P25 for ages 40–49.

Hint. `rng.integers(1, 100, size=10000)` draws the ranks (Lecture 8); draw them after the ages. Store the percentiles in a NumPy array with age groups in rows and P1 to P99 in columns, sorted as numbers so that P2 comes before P10. P1 is then column 0. A loop over the adults with `values[a, r - 1]` works.

2.3 Compare Both Economies with Denmark

Make one table with three rows: observed Denmark, age-group medians only, and variation within age groups. Report P10, P50, P90, P90 − P10 (all in DKK), and the percentage with negative wealth. For Denmark, take P10, P50 and P90 from the all-adult row of FORMUE16 and the negative-wealth percentage from the all-adult rows of FORMUE15.

1. What fraction of the observed P90 − P10 gap does the first economy reproduce? Why does it contain no one with negative wealth?
2. The second economy uses the age-specific percentiles directly. How close should it come to Denmark? Give one reason why a difference remains.

Hint. `np.percentile(wealth, 10)` works like `np.median`, with the percentile as the second argument. `100*np.mean(wealth < 0)` is the percentage below zero.

2.4 An Older Denmark

Keep the age-specific wealth distributions from 2.2, but change the age shares: move 5 percentage points of the adult population from the 18–29 group to the 70–79 group. Run the second economy again with the same seed and report the same five statistics next to your results from 2.2.

1. Does the older population have a higher median? A wider or narrower P90 − P10 gap? More or fewer people with negative wealth? Use your age profiles from Part 1 to explain each answer.
2. Why is it useful to use the same seed in both runs?
3. This experiment holds the 2024 age profiles fixed. Give one reason why they might change if the population really aged.

Hint. If your function takes the age shares as an argument, the new run takes one new line of code.